

# Hadamard 的行列式不等式

Kenneth Lange

**摘要** 本文对 Hadamard (阿达玛) 的行列式不等式提出一个短而初等的证明.

令  $\mathbf{X}$  是一个其元素为复数、列向量为  $\mathbf{x}_1, \dots, \mathbf{x}_n$  的  $n \times n$  矩阵. Hadamard 的行列式不等式 [2] 为

$$|\det \mathbf{X}| \leq \prod_{i=1}^n \|\mathbf{x}_i\|_2.$$

Horn 和 Johnson [1] 对这个经典结果至少提供了 6 个证明. 当某一  $\mathbf{x}_i = \mathbf{0}$  时, Hadamard 不等式显然成立. 否则, 如果在不等式两端同除以其右端, 那么 Hadamard 不等式在 Euclid (欧几里德) 长度限制  $\|\mathbf{x}_i\|_2 = 1$  下就简化为  $|\det \mathbf{X}| \leq 1$ . 取  $\mathbf{X}$  为单位矩阵  $\mathbf{I}$  或任意酉矩阵时就得到一个等式. 回忆一下, 一个酉矩阵有相互垂直的列向量, 因而有

$$|\det \mathbf{X}|^2 = \det \mathbf{X}^* \det \mathbf{X} = \det(\mathbf{X}^* \mathbf{X}) = \det \mathbf{I} = 1.$$

由 Weierstrass (魏尔斯特拉斯) 定理, 在由上述长度限制所定义的紧集上必定达到  $|\det \mathbf{X}|$  的正最大值. 假设  $\mathbf{X}$  的列向量都是单位长度的, 它产生此最大值, 但是其两列, 譬如前两列, 不垂直. 我们来证明: 通过仔细地选择两个复数  $a$  和  $b$ , 用线性组合  $\mathbf{y}_1 = a\mathbf{x}_1 + b\mathbf{x}_2$  代替  $\mathbf{x}_1$ , 值  $|\det \mathbf{X}|$  可以增加. 新矩阵  $\mathbf{Y}$  的行列式满足

$$\det \mathbf{Y} = \det(a\mathbf{x}_1 + b\mathbf{x}_2, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n) = a \det \mathbf{X}.$$

通过提出下述限制条件:

$$|a|^2 + |b|^2 + 2\operatorname{Re}(\bar{a}bs) = 1, \quad s = \mathbf{x}_1^* \mathbf{x}_2,$$

我们迫使  $\mathbf{y}_1$  是一个单位向量. Cauchy-Schwarz (柯西-施瓦茨) 不等式蕴涵着  $|s| \leq 1$ . 此外, 由假设有  $s \neq 0$ , 并且  $|s| \neq 1$ , 因为否则  $\mathbf{x}_1$  和  $\mathbf{x}_2$  共线, 以致  $\det \mathbf{X} = 0$ . 读者可以验证: 我们可以选取

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-|s|^2}} \\ \frac{\bar{s}}{-\sqrt{1-|s|^2}} \end{pmatrix}.$$

由于  $a > 1$ , 恒等式  $\det \mathbf{Y} = a \det \mathbf{X}$  说明了  $|\det \mathbf{Y}| > |\det \mathbf{X}|$ . 这个矛盾证明了  $\mathbf{X}$  的列向量相互正交, 因而  $\mathbf{X}$  是酉矩阵以及  $|\det \mathbf{X}| = 1$ . (下转 38 页)

译自: The Amer. Math. Monthly, Vol. 121 (2014), No. 3, p. 258–259, Hadamard's Determinant Inequality, Kenneth Lange. Copyright ©2014 the Mathematical Association of America. Reprinted with permission. All rights reserved. 美国数学协会授予译文出版许可.

作者的邮箱地址是 [klange@ucla.edu](mailto:klange@ucla.edu).

- [14] J. A. McCammon, B. R. Gelin, and M. Karplus, Dynamics of folded proteins, *Nature*, 267 (1977), 585–590.
- [15] G. Némethy and H. A. Scheraga, Theoretical determination of sterically allowed conformations of a polypeptide chain by a computer method, *Biopolymers*, 3 (1965), 155–184.
- [16] J. C. Phillips et al., Scalable molecular dynamics with NAMD, *J. Comput. Chem.*, 26 (2005), 1781–1802.
- [17] S. Piana, K. Lindorff-Larsen, and D. E. Shaw, Atomic-level description of ubiquitin folding, *Proc. Natl. Acad. Sci. USA*, 110 (2013), 5915–5920.
- [18] A. Rahman and F. H. Stillinger, Molecular dynamics study of liquid water, *J. Chem. Phys.*, 55 (1971), 3336–3359.
- [19] H. A. Scheraga, Respite, adspice, prospice, *Ann. Rev. Biophys.*, 40 (2011), 1–39.
- [20] T. Schlick, *Molecular Modeling: An Interdisciplinary Guide*, 2nd ed., Springer-Verlag, New York, 2010.
- [21] T. Schlick and R. Collepardo-Guevara, Biomolecular modeling and simulation: The productive trajectory of a field, *SIAM News*, 44:6 (2011), 1 and 8.
- [22] T. Schlick, R. Collepardo-Guevara, L. A. Halvorsen, S. Jung, and X. Xiao, Biomolecular modeling and simulation: A field coming of age, *Quart. Rev. Biophys.*, 44 (2011), 191–228.
- [23] Scientific grand challenges: Opportunities in biology at the extreme scale of computing, Workshop Rep., August 2009; [http://science.energy.gov/~media/ascr/pdf/program-documents/docs/Biology\\_report.pdf](http://science.energy.gov/~media/ascr/pdf/program-documents/docs/Biology_report.pdf).
- [24] W. Wang, O. Donini, C. M. Reyes, and P. A. Kollman, Biomolecular simulations: Recent developments in force fields, simulations of enzyme catalysis, protein-ligand, protein-protein, and protein-nucleic acid noncovalent interactions, *Ann. Rev. Biophys. Biomol. Struct.*, 30 (2001), 211–243.
- [25] A. Warshel and M. Karplus, Calculation of ground and excited state potential surfaces of conjugated molecules, I: Formulation and parametrization, *J. Amer. Chem. Soc.*, 94 (1972), 5612–5625.
- [26] A. Warshel and M. Levitt, Theoretical studies of enzymic reactions: Dielectric, electrostatic and steric stabilization of carbonium ion in the reaction of lysozyme, *J. Mol. Biol.*, 103 (1976), 227–249.
- [27] G. Zhao et al., Mature HIV-1 capsid structure by cryo-electron microscopy and all-atom molecular dynamics, *Nature*, 497 (2013), 643–646.

(叶其孝 译    吴庆宝 校)

\*\*\*\*\*

(上接 93 页)

### 参考文献

- [1] R. A. Horn, C. R. Johnson, *Matrix Analysis*, Second edition, Cambridge University Press, Cambridge, 2013.
- [2] J. Hadamard, Résolution d'une question relative aux déterminants, *Bulletin des Sciences Math* 2 (1893) 240–246.

(陆柱家 译    陆昱 校)